

Observation Without a Final Observer: A Record-Based Stopping Condition for Observer Regress in Closed Physical Systems

Yohaku Kimura (Shohei Kimura)

Independent Researcher, Tokyo, Japan

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Abstract

Discussions of quantum measurement often move from a measured system to an apparatus, an environment, an observer, and then to a further observer. That sequence is not by itself an infinite regress. A regress arises only when an additional certification rule is imposed: an observation counts as established only if another observer observes the observer or the resulting record. This article isolates that rule and proposes a Record-Based Closure Principle (RBCP). An observation event is operationally closed when an interaction inside a physically closed system produces a physically instantiated record that is causally correlated with the relevant alternatives. No further observer is constitutive of the earlier event. Environment-induced decoherence explains the effective stability of macroscopic records, while redundant environmental encoding explains a stronger form of public or intersubjective objectivity. Neither mechanism, however, guarantees the truth of a proposition represented by a record or solves the quantum measurement problem by itself. The article distinguishes historical formation closure, present record closure, and public accessibility closure; this three-level model resolves apparent conflicts involving observer extinction, unread records, record destruction, and recoherence. It then extends the analysis to socially reproduced records. Scriptural and other boundary-directed corpora can stabilize an internal reference to an alleged exterior, but they are records of claims and practices, not proofs of a transcendent referent. A series of adversarial tests, including fabricated records, reference-point hijacking, total record loss, and reversal of decoherence, specifies the scope and failure conditions of the proposal.

Keywords: observer regress; quantum measurement; physical records; decoherence; quantum Darwinism; relational quantum mechanics; social records; epistemic closure

1. Introduction

The vocabulary of observation carries an ambiguity that becomes especially visible in foundational discussions of quantum mechanics. A physical system interacts with an apparatus; the apparatus becomes correlated with an environment; an experimenter reads a pointer; another experimenter may inspect the first experimenter's report. This sequence can be extended, and the familiar von Neumann measurement chain gives a formal background for asking where, if anywhere, a cut should be placed (von Neumann, 1955). Yet an extendable chain is not automatically an infinite regress. A regress appears only when every link is made constitutively dependent on certification by a later link.

The central claim of this article is that the alleged need for a final observer is generated by a category mistake. The occurrence of an interaction, the existence of a record, and the justification of a later belief about that interaction are different matters. A further observer may increase evidential confidence, detect error, or make a record publicly accessible. The further observer does not thereby bring the earlier interaction into existence. Once this distinction is enforced, the observer sequence can continue for practical reasons without becoming a constitutive infinity.

To express this point in a form suitable for a closed physical universe, the article develops a Record-Based Closure Principle (RBCP). In its minimal form, RBCP states that an observation event is operationally closed when an internal interaction produces a physical record whose state depends in a discriminating way on the relevant alternatives. The record may be microscopic or macroscopic, local or distributed, currently read or unread. What matters is that the relevant correlation is physically instantiated within the system. A later

reading is a new interaction with its own record; it is not a missing constituent of the first event.

Decoherence and quantum Darwinism supply important but limited resources for this proposal. Decoherence describes how environmental entanglement suppresses locally accessible interference and selects stable pointer structures (Zurek, 2003; Schlosshauer, 2005). Quantum Darwinism describes how information about selected states can be redundantly proliferated into environmental fragments, allowing multiple systems to obtain concordant information without directly disturbing the source (Ollivier et al., 2004; Zurek, 2009). These mechanisms help explain stable and public records. They do not, by themselves, select a unique interpretation of quantum mechanics or derive a single definite outcome. The argument therefore uses decoherence as a theory of effective record stability, not as a concealed solution to the measurement problem.

The proposal also has a second, deliberately separated application. Human societies construct long-lived reference structures through archives, institutions, rituals, technical standards, and scriptural corpora. Such structures can persist, replicate, and cause action. Their existence is a social and physical fact even when the truth of their represented propositions remains unsettled. The article develops a restricted correspondence between physical records and socially reproduced records, while rejecting the stronger claim that social repetition is literally quantum decoherence. In particular, a scriptural text is undeniably a record of a claim about transcendence and of a history of transmission; its mere existence is not a record that reliably tracks the transcendent event asserted.

Five contributions follow. First, the article identifies the precise certification rule that generates the regress. Second, it defines records without requiring a present reader. Third, it distinguishes three temporal and epistemic grades of closure, thereby separating historical occurrence from surviving evidence and public accessibility. Fourth, it constrains the

physical-social analogy through the concept of a boundary-reference record. Fifth, it subjects the framework to adversarial tests rather than protecting it through unrestricted redefinition. The resulting theory is modest: it terminates a specific regress, but it does not guarantee truth, normativity, metaphysical exteriority, or a preferred quantum interpretation.

2. From Measurement Chains to a Certification Regress

A measurement chain is a sequence of interactions and correlations. Let M_0 denote an interaction between a target system and an apparatus. Let M_1 denote a later interaction in which another subsystem reads the apparatus, M_2 a still later interaction in which a report of M_1 is checked, and so on. Nothing in this description requires the chain to be infinite. Actual chains end because interactions cease, records are ignored, resources are finite, or the explanatory purpose has been satisfied. The logical problem begins only when a certification rule is added.

Certification Rule (CR): For any observation event M_n , M_n counts as established only if there exists a later observation M_{n+1} whose object is M_n , its observer, or its record.

CR is recursively unstable. If M_1 is required to establish M_0 , then the same rule requires M_2 to establish M_1 , M_3 to establish M_2 , and so forth. A final observer would stop the series only by being exempted from the rule imposed on every earlier observer. That exemption would be ad hoc unless a different stopping condition were supplied. The regress is therefore not solved by discovering a privileged last witness; it is dissolved by rejecting CR as a constitutive condition of observation.

Three predicates must be separated. Occurrence concerns whether an interaction took place. Record formation concerns whether the interaction left a physically instantiated difference. Warrant concerns whether an agent is justified in believing a proposition about the interaction. These predicates can come apart. An event may occur and leave a record that no present agent knows exists. A record may exist but be corrupted, misleading, or badly

interpreted. An agent may be justified by convergent evidence even when no single record is decisive. The move from occurrence to warrant is epistemic; treating that move as a condition for occurrence produces the regress.

The distinction can be stated as a non-identity thesis: establishing that p for an agent is not identical to making p the case. This thesis is compatible with ordinary scientific practice. Instrument logs, detector tracks, photographic plates, server records, and memory states are scrutinized because they can fail. Scrutiny presupposes that the candidate record already exists as an object of investigation. A second observation can certify, disconfirm, or reinterpret the first record, but it cannot be required for the first record to possess a physical state.

The target of RBCP is accordingly narrower than the quantum measurement problem. The measurement problem asks, among other things, how a theory with linear quantum dynamics is related to definite outcomes and classical appearances. Decoherence is central to that discussion but is not by itself an interpretation or a collapse mechanism (Schlosshauer, 2005). RBCP instead addresses a philosophical regress generated by demanding that every observation be validated by another observer. A theory may accept RBCP while remaining undecided about collapse, branching, relational facts, or agent-centered probabilities.

This scope restriction matters. It prevents a local solution from masquerading as a universal one. The article does not claim that physicists generally endorse CR, nor that standard quantum mechanics formally contains an infinite observer sequence. It claims that whenever CR is invoked - explicitly or implicitly - the resulting regress can be stopped by shifting the constitutive condition from further observation to internal record formation.

3. Internal Record Formation in a Closed Physical System

Consider a universe U modeled as a closed physical system. 'Closed' here means that the dynamics used in the model do not require external inputs or degrees of freedom. The

assumption does not decide whether U is spatially finite, temporally bounded, or represented by a universal wave function. It states only that the correlations relevant to observation can be described through interactions among subsystems of U . No standpoint outside U is needed to make the model's variables meaningful.

For an idealized measurement interaction, decompose the relevant degrees of freedom into a target system S , an apparatus A , and an environment E . An initial state may be represented schematically as

$$|\Psi_0\rangle = (\sum_i c_i |s_i\rangle)|a_0\rangle|e_0\rangle,$$

and an interaction correlates the alternatives according to

$$|\Psi\rangle = \sum_i c_i |s_i\rangle|a_i\rangle|e_i\rangle.$$

The notation is intentionally generic. It does not assert that the alternatives $|s_i\rangle$ possess absolute, observer-independent values before interaction, nor does it impose a collapse postulate. It represents a correlation structure internal to U . This structure can be interpreted in different ways: as a relative state in an Everettian setting (Everett, 1957), as facts relative to interacting systems in relational quantum mechanics (Rovelli, 1996), or as part of an operational account in which the apparatus is treated classically at the relevant stage.

Environmental entanglement changes which coherences are locally accessible. When the environmental states associated with distinct apparatus states become nearly orthogonal, the off-diagonal terms of the reduced state of S and A are strongly suppressed. In suitable regimes, a preferred set of pointer states is selected by the system-environment interaction, and those states support robust records (Zurek, 2003). The result is effective classicality for local purposes, not an assertion that the global state has undergone a fundamental nonunitary collapse.

A physical record can therefore be defined through correlation and distinguishability rather than through conscious awareness. For each relevant alternative, consider the

conditional state of a candidate record-bearing subsystem R . R functions as a record when those conditional states remain operationally distinguishable above a specified threshold for the timescale and coarse-graining relevant to the inquiry. The threshold is contextual because every actual record has finite resolution, finite lifetime, and finite resistance to noise. This contextuality does not make the record subjective; it makes the claim about what is recorded explicitly scale-relative.

Three minimal conditions follow. First, the record must be causally downstream of the interaction: had the relevant alternative differed, the record state would have differed across the selected partition. Second, the difference must be physically instantiated somewhere in U , whether in an apparatus pointer, a photon field, a memory state, a written mark, or a distributed data structure. Third, the difference must persist long enough to function as a record under the inquiry's temporal resolution. No actual reader is included in these minimal conditions.

Redundancy introduces a stronger condition. If multiple disjoint fragments of E carry approximately the same information about a selected observable, then several systems can independently recover concordant information by interacting with different fragments. This is the central operational idea of environment as witness and quantum Darwinism (Ollivier et al., 2004; Zurek, 2009). Redundancy does not turn a record into an infallible statement. It makes the record resilient, multiply accessible, and capable of supporting intersubjective agreement.

The observer can now be minimally physicalized. An observer is any subsystem capable of becoming correlated with a record and retaining a state change that can enter later dynamics. Consciousness, personhood, and linguistic report are unnecessary at this level. This position is close to the relational insistence that physical systems stand on both sides of an interaction rather than being divided into a fundamentally quantum object and a metaphysically exceptional subject (Rovelli, 1996). Di Biagio and Rovelli (2021) further

distinguish relative facts from stable facts whose relativity can be ignored within a decoherence domain; that distinction provides a useful comparison for the present account of stable records.

The account remains more record-realist than QBism. QBism treats quantum theory as a normative tool used by an agent to organize expectations about personal experience (Fuchs et al., 2014). RBCP, by contrast, permits a record-bearing subsystem to exist without being experienced by a current agent. This difference should not be concealed under the label of interpretive neutrality. The proposal is better described as interpretation-flexible: it can be embedded in several physical interpretations, but it does not preserve every interpretation without modification.

4. The Record-Based Closure Principle

Definition 1 (Observation record). A physical state or distributed set of states R within U is an observation record of event e , relative to a specified partition and coarse-graining, when (i) R is causally correlated with the relevant alternatives of e ; (ii) the correlated alternatives are physically distinguishable above a stated threshold; and (iii) that distinguishability persists for a stated timescale. Current readability, current knowledge of R 's location, and current linguistic interpretation are not necessary conditions.

Definition 2 (Record-Based Closure Principle). If an interaction event e produces an observation record R within U , then the occurrence of e as an observation event does not require a further observer to observe R . A later interaction with R is a new event and may generate a new record, but it is not constitutive of the earlier event.

RBCP is a criterion of constitutive sufficiency and event individuation, not an additional dynamical law. Its empirical application depends on whether the stated record conditions are physically satisfied in a particular model or experiment.

RBCP blocks CR at its first iteration. Suppose M_0 produces R_0 . CR demands M_1 to establish M_0 . RBCP permits M_1 , but denies that M_1 is required for M_0 to have occurred. If M_1 takes place, it produces its own record R_1 , and a later system may inspect R_1 . The sequence can continue indefinitely in principle, but each link is complete under its own internal record condition. The series is open-ended without being regressively constitutive.

Proposition 1 (Non-constitutive certification). An observation of a record can alter the warrant attached to an earlier observation without altering the constitutive conditions under which the earlier observation occurred. The proof is conceptual. The later observation has different interacting systems, a later time, and a distinct causal product. Treating it as part of the earlier event collapses two events into one and thereby manufactures the dependence that drives the regress.

The principle depends on a minimal realism about physical states: a record is not brought into existence by being known. That realism is already presupposed whenever a physical theory describes unattended detectors, unobserved environmental fragments, or data stored before retrieval. To say that R exists is not to say that an agent can prove that it exists. The existence claim belongs to the model; the proof claim belongs to an epistemic situation within the model.

Unread and inaccessible records illustrate the distinction. An encrypted archive can remain a record of a prior encoding process even when no current agent possesses the key. A distributed record can survive in fragments even when no single subsystem can reconstruct the whole. A geological trace can record a causal history before any organism is capable of interpreting it. In each case, whether the object records a specific event depends on its causal relation to that event, not on the presence of a current reader. Agents may nevertheless be unable to determine whether the relation obtains.

This last point motivates an epistemic non-equivalence: unknown is not identical to absent. If R exists but no agent can locate it, the closure condition may obtain while remaining unknown. If R does not exist but agents believe that it does, their belief does not create closure. If available evidence underdetermines R 's existence, the agents face an epistemic suspension; the model does not need a third physical value between existence and nonexistence at the effective record level.

Definition 3 (Record of an event versus record of a claim). A material inscription that states p is, at minimum, a record that p was inscribed, asserted, transmitted, or stored. It is a record of the event p only when an appropriate causal tracking relation connects p to the inscription. This distinction is indispensable. A forged laboratory log is a real record of forgery and data entry; it may fail to record the experimental result it purports to represent. A mythic narrative is a real record of narrative production and social transmission; it is not, merely by surviving, reliable evidence that every event it describes occurred.

The truth-neutrality of RBCP is therefore precise rather than indiscriminate. RBCP does not declare all record contents equally true. It declares that truth certification is not part of the minimal condition by which an observation event avoids dependence on another observer. Reliability, authenticity, semantic content, and justification remain legitimate questions at a different analytical level.

A useful analogy comes from information theory. Shannon information concerns distinctions among possible signals and is deliberately neutral about meaning (Shannon, 1948). Likewise, the minimal record relation concerns a discriminating physical correlation. Semantic interpretation can be layered onto that relation, but it should not be imported into the physical definition without argument. The record condition is thus thin enough to stop the observer regress and thick enough to exclude arbitrary objects that bear no causal relation to the event.

5. Three Grades of Closure and Their Temporal Logic

A single use of the word 'closure' obscures differences that become decisive when observers die, records decay, or public access disappears. The framework therefore distinguishes three grades.

Table 1. Three grades of record-based closure

Grade	Condition	What it secures	What can remove it
Historical formation closure	A record-forming interaction occurred at t_1 .	The event was operationally complete under the record criterion at t_1 .	Only a genuine reversal that removes the relevant event structure; ordinary later loss does not retroactively cancel it.
Present record closure	At least one relevant record survives at time t .	A current physical stopping structure exists, whether read or unread.	Destruction, overwriting, decay, or loss of every relevant record.
Public accessibility closure	Redundant or independently accessible records support concordant retrieval.	Intersubjective or multi-system agreement can be operationally produced.	Loss of redundancy, readers, access, or agreement-producing channels.

Note. The grades are cumulative in ordinary cases but analytically separable.

Historical formation closure concerns an event at the time of record production. Once an interaction has generated a record under the adopted effective description, the later death of the observer or ordinary destruction of the record does not make it false that record formation occurred. This is a claim about temporal non-retroactivity, not about eternal preservation.

Present record closure concerns whether at least one relevant record still exists at a later time. It can be lost. A server may be wiped, a manuscript burned, a memory overwritten, or all environmental traces dispersed beyond recovery. The loss removes a current stopping structure for agents attempting to reconstruct the event, even though it does not ordinarily reverse the prior causal history.

Public accessibility closure is stronger again. It requires sufficiently redundant or independently accessible records to support concordant retrieval by multiple subsystems. This grade explains why some facts become publicly stable while others remain private, fragile, or contested. Quantum Darwinism addresses an analogous operational condition at the physical level: multiple observers can consult different environmental fragments and agree about selected pointer information (Ollivier et al., 2004).

Proposition 2 (Non-retroactivity under ordinary loss). If record formation occurred at t_1 and a surviving record is destroyed at t_2 , t_2 eliminates present record closure but does not, by that fact alone, negate historical formation closure. The qualification 'ordinary' is essential. A complete physical reversal that removes all correlations and restores interference is not equivalent to burning a document or deleting a local file; it raises separate issues about whether the earlier event remains attributable as a stable fact.

The distinction also repairs an apparent conflict about action. A record need not currently cause behavior in order to remain a record. Actual action connection is a criterion for active social or operational use, not for minimal physical recordhood. It is enough that the record is a causally structured state produced by the event. The capacity for future coupling may be absent or unknown; the historical relation can still obtain.

These grades prevent two opposite errors. The first error makes observation depend on an immortal community of readers. The second makes every past record eternally available merely because it once existed. The three-grade model allows historical occurrence to persist as a temporal fact, present evidence to disappear, and public objectivity to rise or fall with redundancy and access.

6. Physical Records and Social Reference Structures

Physical records and social records share a structural pattern: an event produces a discriminating mark, the mark persists or is reproduced, and later systems can couple to it. The similarity is useful, but it is not an identity. Quantum decoherence concerns the suppression of locally accessible phase relations through environmental entanglement. Social stabilization concerns copying, education, institutional enforcement, ritual repetition, archival practice, and technological storage. Calling both processes 'decoherence' without qualification would replace explanation with metaphor.

The defensible claim is a partial structural correspondence. Physical correlation corresponds to the production of an inscription or memory. Physical persistence corresponds to storage, maintenance, and recopying. Redundant environmental encoding corresponds, at a higher descriptive level, to multiple archives, editions, witnesses, and transmission channels. Downstream coupling corresponds to actions, sanctions, institutional procedures, or new records caused by earlier records. The mechanisms differ, but the abstract sequence - generation, stabilization, redundancy, and causal reuse - can be compared.

Social objectivity is also operational rather than metaphysically absolute. A rule, currency, office, or classification can become publicly stable because many agents and institutions reproduce the same recognition conditions. Searle's analysis of institutional facts emphasizes that collective practices can create objective social consequences without converting those consequences into observer-independent natural kinds (Searle, 1995). The present account adds that material records are among the mechanisms by which such practices become temporally extended and multiply accessible.

Scriptural corpora provide a particularly demanding case because they contain claims about divine, transcendent, or universe-external agency. Define a boundary-reference record B as an internal record that represents a relation between U and an alleged domain, agent, or cause not contained within the ordinary descriptive boundary of U. B can be copied, interpreted, institutionalized, and made causally efficacious inside U. In that limited sense, B stabilizes an internal reference to an alleged exterior.

The existence of B establishes less than the represented proposition. It establishes that a boundary claim exists in U, that the claim has a material history, and perhaps that it has caused durable social effects. It does not establish that the alleged exterior exists, that a transcendent observer has observed U, or that the text reliably records communication from such an observer. B is a record of a claim unless an independent causal tracking relation can

be shown. This distinction converts the scriptural example from an ontological proof into a theory of reference stabilization.

Boundary-reference closure is therefore semantic and institutional. Internal agents need not search for an ever more external text that records the claim; the reference is materially present in B. But this semantic closure should not be confused with the physical stopping condition for observation. The physical regress has already been blocked by RBCP at the level of internal records. B adds a special case in which the content of a record points beyond the system's ordinary conceptual boundary.

A civilization without scripture does not threaten RBCP. Detector traces, memories, archives, technical standards, and legal documents can all instantiate record structures without containing claims about a transcendent exterior. What disappears in a scripture-free civilization is not physical closure, but one particular form of boundary-reference closure. This result prevents the theory from depending on the historical contingency of terrestrial religion.

The same restriction handles normativity. A record can shape law, ethics, education, or collective identity without becoming true by virtue of its efficacy. Causal success and normative authority are different from descriptive truth. The framework studies how a reference structure becomes stable and action-guiding; it does not infer that stability confers correctness.

7. Adversarial Tests

The proposal is best evaluated through destructive tests rather than by accumulating favorable examples. The methodological aim resembles Popper's emphasis on risky criticism and Lakatos's concern with whether a research programme responds to anomalies by clarification or by ad hoc insulation (Popper, 1959; Lakatos, 1978). The following tests identify what the theory preserves, what it concedes, and where it fails.

7.1 Scripture-Free Civilization

Remove every religious scripture from a civilization while leaving ordinary physical interactions, instruments, memories, archives, and institutions intact. Observation records still form; decoherence still stabilizes suitable macroscopic degrees of freedom; redundant records can still support public agreement. RBCP therefore survives. Only the special case of boundary-reference closure through scripture is absent. The test shows that scripture is an application, not a necessary premise of physical closure.

7.2 Fabricated Records and Reference-Point Hijacking

Suppose an attacker fabricates a claim of contact with an exterior agent, distributes it across media, embeds it in education and institutions, and causes generations of action. The fabrication may satisfy conditions of persistence, redundancy, and social efficacy. The test therefore defeats any theory that equates redundancy with truth. It does not defeat RBCP, because RBCP predicts precisely that a fabricated record can be a real record of fabrication and can become a powerful social reference point.

The distinction between a generic record and a stopping reference point must nevertheless be retained. A viral image or popular novel is not automatically a boundary-reference record. For that role, the record's content must represent a relation to an alleged exterior, and the record must participate in a durable reference network. Even then, its role is semantic and institutional, not evidentially self-validating. If an attacker succeeds completely, the result is a new stable social formation, not a proof of the claim.

7.3 Plural and Mutually Inconsistent Stopping Points

Multiple incompatible records can be stable at the same time. Different communities may preserve contradictory boundary claims; different apparatuses may contain inconsistent data because of error or noise. RBCP does not require a unique global record. Minimal closure is local to the event and record relation under analysis. Public objectivity requires additional

convergence conditions, and truth requires still further evaluation. Plurality therefore weakens consensus without reintroducing the constitutive observer regress.

7.4 Static Persistence Versus Reproduction

No material carrier is eternal. Thermal noise, decay, damage, editing, and erasure threaten every record. The theory fails if persistence is defined as an immutable inscription preserved without maintenance. A more defensible account treats long-term persistence as a property of a distributed and sometimes self-reproducing record structure. Copies replace damaged copies; error correction preserves distinctions; institutions train new readers; environmental fragments provide multiple physical carriers.

This dynamic account is consistent with decoherence. Decoherence does not preserve a symbol forever; it helps select stable macroscopic alternatives by dispersing phase information into the environment. The subsequent durability of a document, memory, or instrument reading depends on material and informational maintenance. Landauer's principle further reminds us that information processing and erasure are physical operations rather than costless abstractions (Landauer, 1961).

7.5 Observer Extinction and Loss of Action Connection

Suppose every biological observer disappears while records remain. A preliminary, action-centered definition might conclude that observation itself has ceased because no subsystem reads the records and acts on them. That conclusion is too strong. Actual downstream action is necessary for an actively used social reference network, but not for the historical fact that a record-forming interaction occurred or for the present physical existence of the record.

The three-grade model resolves the case. Historical formation closure remains because the interaction and record formation occurred. Present record closure remains if the records survive. Public accessibility closure disappears because no independent readers are

available. If a later observer evolves or arrives, its reading initiates a new interaction and may restore public accessibility; it does not retroactively create the earlier record.

7.6 Total Record Annihilation

Now suppose every surviving copy and physical trace of a record is destroyed. Present record closure and public accessibility closure are lost. Historical formation closure remains within the effective causal history: the fact that a record once formed is not made false by ordinary later destruction. However, if no trace survives, later agents may be unable to establish that history. The theory therefore concedes a loss of current evidential structure without converting evidential loss into retroactive nonoccurrence.

The same analysis applies to boundary-reference records. Eliminating every relevant corpus would eliminate the present internal reference structure, even though it would not make the former acts of inscription and belief never to have occurred. Proving that every corpus was humanly authored would establish facts about production; it would not, by itself, prove the falsity of every proposition about an exterior. Conversely, the inability to prove falsity does not confer truth.

7.7 Recoherence and Reversal

Quantum dynamics is not fundamentally irreversible merely because decoherence is practically irreversible. In sufficiently controlled systems, interference can revive, and information dispersed into an environment can in principle be partially recovered. RBCP must therefore avoid treating every record as absolutely indelible. Its claim is indexed to a stability regime: ordinary resources, a specified coarse-graining, and a timescale over which the record alternatives remain distinguishable.

A local erasure that destroys a memory while leaving environmental and downstream traces is not a reversal of the entire record-forming interaction. It removes some surviving records but does not restore the full premeasurement state. A genuine global reversal that

eliminates every correlation and revives interference would be a stronger case. In such a regime, attributing a unique, stable past record may become interpretation-dependent. RBCP does not claim authority beyond the effective record domain; the domain may shrink when extraordinary control restores coherence.

7.8 The Test Network

The tests constrain one another. Generalizing from scripture to records prevents historical contingency, but fabrication tests prevent that generalization from becoming a truth-free license. Dynamic persistence prevents dependence on an eternal carrier, while observer-extinction tests prevent active readership from being smuggled back into the minimal definition. Reversal tests prevent practical irreversibility from becoming a metaphysical absolute. The framework survives only by preserving these paired constraints.

This network has the character of a research programme's protective structure, but its function is not to immunize the core from criticism. Each test identifies an explicit failure condition. The theory fails as a truth theory if it equates records with truth; it fails as a universal physical account if it claims that decoherence solves definite outcomes; it fails as a social theory if it treats repetition as literal quantum dynamics; and it fails as a temporal theory if ordinary record loss is confused with exact reversal or retroactive cancellation.

8. Relation to Quantum Interpretations and the Measurement Problem

RBCP occupies an intermediate level between bare formalism and full interpretation. It requires enough physical structure to speak of interactions, correlations, records, and effective stability. It does not specify whether the universal dynamics collapse, branch, remain perspectival, or function as an agent's normative calculus. The resulting position is constrained pluralism rather than interpretive neutrality.

In Copenhagen-style practice, RBCP can be read as an explicit account of why an apparatus record is a legitimate stopping point for operational description. Bohr's insistence

on the conditions under which a phenomenon can be unambiguously communicated already directs attention toward stable experimental arrangements rather than an infinite hierarchy of minds (Bohr, 1928). RBCP removes consciousness from the stopping rule and locates closure in the record-producing interaction.

In Everettian approaches, the global interaction remains unitary, and records are branch-relative correlations within the entangled state (Everett, 1957). RBCP is compatible with that description provided closure is understood relative to a decohered record structure rather than as a unique global collapse. The principle does not decide whether all branches are equally real or how probability should be understood.

Relational quantum mechanics offers a close conceptual neighbor. It treats values as relative to interacting systems and rejects a privileged observer-observed divide (Rovelli, 1996). The distinction between relative facts and stable facts further shows how decoherence can make relational labels negligible within a domain (Di Biagio & Rovelli, 2021). RBCP can be formulated relationally by indexing a record to the systems and coarse-graining under which its alternatives are distinguishable. It should not be read as an absolute God's-eye inventory of records.

QBism marks the clearest boundary. Because QBism centers an agent's experiences and personal probabilities, it resists treating an unread physical state as an observation in the same sense used here (Fuchs et al., 2014). A QBist can still accept that a later experience does not retroactively create an earlier experience, but the agent-independent record realism of RBCP would require translation. This divergence is not a defect; it identifies a substantive assumption rather than hiding it.

Most importantly, decoherence does not supply a unique-outcome postulate. The reduced state may become approximately diagonal while the global state remains a superposition. Schlosshauer (2005) emphasizes that decoherence must be interpreted within a

broader account of quantum mechanics. RBCP uses the diagonalized record structure to explain why further observers are unnecessary for operational closure. It does not infer from that structure that only one outcome exists in every interpretation.

The proposal is also compatible with the possibility that the language of 'observation' is dispensable. One could replace it with 'record-producing interaction' throughout. The philosophical value of retaining the term is diagnostic: it shows that the regress arises from importing epistemic certification into a physical relation. Once the relation is stated physically, the final observer ceases to be a required component.

9. Objections and Replies

Objection 1: RBCP merely stipulates that the regress stops at a record. Reply: The principle is indeed a stopping condition, but it is not arbitrary. It follows from distinguishing two events that CR conflates: the original record-producing interaction and a later inspection of that record. Requiring the later event to constitute the earlier one reverses the causal order. RBCP restores the ordinary causal individuation of events.

Objection 2: An unobserved record cannot be known to exist, so it cannot stop anything. Reply: This objection again replaces an ontic condition with an epistemic one. An agent may not know whether closure obtains. The proposal does not promise universal decidability. It states that if the record exists, no additional observer is needed for its existence. Unknown closure is still different from an infinite chain of required observers.

Objection 3: A record can be false, forged, or hallucinatory. Reply: A record's existence and the truth of its represented proposition are separate. A forged record reliably records at least some causal history - its production, storage, and transmission - even if it misrepresents the target event. RBCP is a theory of constitutive closure, not a theory of evidential reliability. Treating it as the latter would be a scope error.

Objection 4: Physical information does not possess meaning without an interpreter.

Reply: The minimal record relation is not semantic. It is a causal and counterfactual dependence among distinguishable states. Meaning enters when a system uses those distinctions under a code, practice, or interpretive scheme. The layered account deliberately separates physical recordhood from semantic content, while allowing semantic records to supervene on physical carriers.

Objection 5: The physical-social mapping is a category mistake. Reply: It would be a category mistake if social replication were identified with quantum decoherence. The article rejects that identity. The claim is only that both domains can instantiate an abstract pattern of record generation, stabilization, redundancy, and downstream causal use. Differences in mechanism and explanatory level remain explicit.

Objection 6: A scriptural record cannot close an observer regress because it is merely a text. Reply: Correct with respect to physical measurement. The scriptural case supplies only boundary-reference closure: an internal, materially stabilized representation of an alleged exterior. It neither proves the exterior nor replaces the physical record condition. The distinction is a limitation of the proposal and a repair of any stronger theological reading.

Objection 7: Claims about a universe-external agent may be empirically testable if that agent produces internal effects. Reply: The framework does not deny partial testability. If an alleged exterior has stable, discriminating internal consequences, those consequences can be investigated as records. What cannot be assumed is that the existence of a text alone establishes the causal source it names. The degree of testability depends on the causal accessibility built into the hypothesis.

Objection 8: Physics does not need a final conscious observer in the first place. Reply: This is largely correct and reinforces the article's scope. RBCP is not offered as a repair to every formulation of quantum theory. It is a diagnosis of a recurrent philosophical demand

for certification by a further observer. Where that demand is absent, RBCP makes explicit a closure condition already implicit in record-centered practice.

Objection 9: If every record is indexed to a partition, threshold, and timescale, closure becomes arbitrary. Reply: Context sensitivity is not arbitrariness when the parameters are stated and constrained by the physical task. A bubble-chamber track, a qubit memory, and a geological stratum operate at different resolutions. Demanding one absolute threshold for all records would erase the very conditions under which records function. The relevant question is whether the chosen parameters are independently motivated and whether the result is robust under reasonable variation.

10. Implications and Research Programme

The record-based approach changes the grammar of the observer question. Instead of asking who finally sees the observer, it asks which interaction established a discriminating physical record, how long that record remains stable, and whether it is redundantly available. These are finite questions about correlations, dynamics, and access. A later observer can answer them imperfectly, but the later observer is not inserted into the earlier event as a metaphysical component.

For the philosophy of measurement, the principal implication is a separation of tasks. Decoherence and redundancy explain why certain records are stable and publicly recoverable. An interpretation of quantum mechanics explains what those records imply about outcomes, states, or worlds. Epistemology evaluates the reliability of inferences from records. Social theory explains how records become institutionalized. The observer regress emerges when these tasks are compressed into one demand for a final certifier.

For epistemology, RBCP offers neither foundational certainty nor an infallible given. A chain of testimony or documentation may rationally terminate in records because records are causally structured evidence, not because they are immune to error. Further checking

remains valuable, but checking is defeasible inquiry rather than an ontological prerequisite. This position resembles a naturalized evidentialism more than a Cartesian foundation.

For social ontology, the distinction between active and dormant reference is especially useful. A record may remain physically present after the community that used it disappears. Its historical and present record closure can survive while its public and normative closure collapse. If a later community decodes the record, a new action-connected reference network begins; the later network does not retroactively create the earlier inscription.

Several formal extensions are available. First, record relations can be represented as directed graphs in which nodes are events or record-bearing subsystems and edges encode causal discrimination, copying, or reading. The regress-generating rule would require every node to have an outgoing certification edge; RBCP instead allows terminal record nodes relative to a stated inquiry. Second, robustness can be quantified by distinguishability thresholds, redundancy measures, and expected survival times. Third, adversarial models can separate redundancy from reliability by allowing coordinated fabrication, correlated error, and institutional capture.

A further extension should examine multi-agent quantum scenarios. In Wigner-type settings, different agents may assign different states or facts to the same laboratory. RBCP predicts that closure must be indexed to the relevant interaction, record, and decoherence domain rather than imposed globally. Whether this indexing can coexist with specific no-go results is a substantive question for future work, not something settled by the present conceptual principle.

The framework's strongest claim is correspondingly limited. An observation does not require an observer of the observer when a record-producing interaction has already occurred within the physical system. The framework's strongest refusal is equally important: stable reference is not truth, redundancy is not justification, social efficacy is not normativity, and a

boundary-directed text is not proof of an exterior. By holding those separations fixed, the theory stops the regress without inventing a final eye and without turning the stopping point into an oracle.

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Author contributions: The sole author conceived the argument, developed the framework, and approved the manuscript.

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